Pivotum

THE DEGREE RISK INDEX · FALL 2026 EDITION

So you're thinking about cybersecurity

The short version — written for you, not about you.

Everything that matters, none of the bits written for your parents.

You've probably seen the number: **4.8 million unfilled cybersecurity jobs worldwide.**

It's real. It's also being sold to you in a way that leaves out the part that matters.

We scored twenty-seven careers on AI exposure. Incident response and threat hunting came out at **5.0 out of 10**. Tier-one SOC analyst work — the exact role most people are told to start in — came out at 7.4.

The 60-second version

- Cybersecurity is the best-scoring career in technology. That's true.
- But its *best* score is 5.0. Licensed engineering is 4.0. A service electrician is 2.5.
- **The 4.8 million shortage is a shortage of *experience*, not of people.**
- Gartner projects over half of tier-one SOC work will be AI-handled by 2028.
- And there's something you can start building tonight, free, that matters more than any degree.

What protects it — and this one's genuinely unique

Every other protected career in our index is protected by something structural: a licence, physical presence, legal liability.

Cybersecurity has almost none of that. No licence. Fully digital. And it still holds at 5.0.

Here's why.

AI is good at patterns and bad at genuine novelty. Most careers get a high novelty score because the work is *complex* — a weird patient, an unusual building site.

Cybersecurity gets it because someone is deliberately manufacturing novelty against you.

An unusual medical case is unusual by accident. An unusual attack is unusual **on purpose** — built specifically to look normal, or to exploit the gap between what a detection system was trained on and what's actually happening.

Which creates something nothing else in our index has: the novelty is *adaptive*. Whatever pattern a system learns to spot, the other side is paid to stop matching it. There's no version of this where the patterns settle down and the job gets routine.

The general lesson worth carrying: an adversary protects a job better than complexity does. Complexity can be learned. An opponent who changes when you learn them can't.

But look where that protection stops

A tier-one SOC analyst isn't facing an adversary. They're working an alert queue against a playbook — pattern-matching, in a role explicitly designed to be pattern-matching.

That's why it scores 7.4 while threat hunting scores 5.0.

ROLE	SCORE
Incident response / threat hunting	5.0
Security architecture	5.5
Penetration testing	5.7
GRC / compliance	6.1
SOC analyst (tier 1)	7.4

About that 4.8 million number

The figures are all real:

- **4.8 million** unfilled positions globally (ISC2)
- **~700,000–750,000** unfilled in the US (CyberSeek — CompTIA, Lightcast and NIST)
- BLS projects **29–33% growth** through 2034, about seven times average

Now read them carefully, because bootcamps and universities won't.

The shortage is for experience, not for people. Practitioners consistently report that "entry level" postings ask for two years of experience. Companies want people who can direct security work, not people who need training. The gap persists *because* new entrants aren't being developed.

Budget has overtaken talent as the real barrier. ISC2's 2025 study found budget cuts displacing skills shortage as the top hiring obstacle. A vacancy with no funded headcount behind it doesn't hire anyone.

And the route you're being sold is the one being automated. Gartner projects more than 50% of tier-one SOC responsibilities handled by AI by 2028 — which is precisely the job every certification and bootcamp points you toward.

None of that means don't do it. It means don't do it the way it's being marketed.

What we won't soften

The burnout data is unusually well documented, and it's bad.

- Bitsight surveyed 1,000+ security professionals: **47% reported some level of burnout**
- Proofpoint found **63% of CISOs** had experienced or witnessed burnout in a year
- Sophos measured **4.8 hours a week lost to burnout** — up over 25% year on year
- Splunk found **52% of security teams overworked**
- Attrition runs higher than other tech roles

On-call is permanent, not a starter phase. Incidents happen at 2am and on Saturdays.

And the certification treadmill never stops, much of it self-funded.

The thing you can do tonight

This is the most useful sentence in this document.

Build hands-on evidence. Home lab, capture-the-flag competitions, bug bounties, contributing to open-source security tools.

In a field with **no licence**, demonstrable capability is the credential. And unlike almost every other career we've looked at, it's completely free and available to you right now.

It also solves the experience paradox — the one thing standing between you and a first job. And it answers the question no amount of thinking will: whether you actually enjoy this when nobody is making you do it.

The entry routes are unusually open to people who've done this. ISC2's research found around 90% of organisations will consider candidates with only IT experience, and 89% accept entry certifications instead of a degree.

Very few fields are that open. But they want evidence, not enthusiasm.

Where to actually aim

Incident response and threat hunting — 5.0. Facing the adversary directly. Best score.

Security architecture — 5.5. System-level judgment where being wrong is expensive.

Offensive security / pen testing — 5.7. Creative adversarial work.

Or the emerging specialisms — cloud security, application security, and **AI/ML defence**, which barely existed three years ago and is named consistently among the most in-demand skills.

Take a SOC role if it's your way in — but treat it as a two-year staging post, not a career. Be deliberate about moving up.

Does this actually sound like you?

The defining trait: do you instinctively ask how something could be **abused**, rather than how it works?

That's largely untrainable and it shows up early. If you've ever looked at a system and immediately thought about how to break it, that's the instinct.

It also suits people who **stay calm when things are going badly** — incidents *are* the job — who are comfortable **deciding without complete information**, and who **enjoy** continuous learning rather than tolerating it.

Harder if you need predictable hours, want tasks that get finished, or picked it because it was marketed as the safe tech career. That last one is what half this document is about.

What actually matters when picking a course

- **Lab hours.** How much time in real environments, actually breaking and defending things?
- **Adversarial thinking, or tool training?** A curriculum organised around specific products is training you for the automating tier.
- **Specialisation pathways** — IR, offensive, cloud, appsec. Generic security degrees place worst.

- **Is there an apprenticeship route?** It solves the experience problem directly and almost nobody uses them.
- **What job titles do graduates have at six months** — SOC, or engineering?

And seriously consider computer science with a security specialisation instead of a cybersecurity degree. The engineering foundation is what verification actually rests on.

Two things worth doing

Start a home lab this month. Genuinely — it costs nothing, it's the strongest signal you can send an employer, and it will tell you within a few weeks whether you like this.

Then find someone two or three years into a security job and ask what they actually do all day — and how often they get woken up.

One last thing

Cybersecurity is a good career. It's just not the frictionless six-figure on-ramp the marketing suggests, and the specific route being sold to you is the weakest one.

What's genuinely true: there's a real opponent, which means the work never gets boring — and you can start building verifiable skill tonight, for free, before you've applied anywhere.

That's a better reason to do this than any workforce statistic.

If you want to go further — three things worth twenty minutes

- **Information Security Analysts — Occupational Outlook Handbook** — the government's own growth and pay data. Nobody's selling you anything.
- **CyberSeek** — an interactive map of actual US demand by role and location. Look up what's genuinely open near you rather than the global aggregate.
- **ISC2 Workforce Study** — where the 4.8 million comes from. Read the actual study rather than the headline; the 2025 finding about budget constraints is the part nobody quotes.

There's a longer version behind this — full scores by role, how they were worked out, the burnout data in detail, what to ask a program, and where we might be wrong. Your parents have it.